What FP4 Costs Long Reasoning on Blackwell, Measured Above the Noise

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Abstract

Blackwell GPUs run NVFP4 natively, but reasoning models must stay accurate over tens of thousands of generated tokens, and the single-checkpoint, single-benchmark comparisons that rank quantization methods are noisier than the gaps they report: on AIME, one GPTQ checkpoint scores 6.9 points apart in two four-sample runs that differ only in the sampling seed, and at four samples per problem a paired comparison over 90 problems resolves only ±4.7 points. We measure Qwen3-1.7B, 8B and 14B on the kernels vLLM deploys on an RTX 5090 (SM120) and a B200 (SM100), with repeated calibration draws, paired tests and exact accuracy-versus-budget curves. The effects that hold up concern length and memory more than accuracy. Calibrating GPTQ on reasoning traces removes most of the extra length that chat calibration adds to Qwen3, in every draw on both GPUs; no published NVFP4 recipe beats trace-calibrated GPTQ, and few accuracy differences between 8B FP4 checkpoints survive multiple-comparison correction. Even trace-calibrated, 8B W4A4 is −2.6 [−4.8, −0.4] points from BF16 over both GPUs' samples, where FP8 with an FP8 cache is +0.2, and the loss is in the weights: activation error is uniform over tokens and, on trace-calibrated weights, equivalent to none within 2.4 points, as is an FP8 KV cache within 2.8, while NVFP4 KV collapses the 1.7B model, as its teacher-forced error predicts. On a 32 GB card long-trace serving is bound by KV-cache capacity, and a capacity model built on it predicts the serving rate, also when fitted on 8B and applied to 14B. At 8B, FP4 weights give only 1.09× the correct answers per minute of FP8 with an FP8 cache on the RTX 5090's samples and 1.05× at the accuracies of both GPUs, and Qwen3-14B, which FP4 fits at the rate of 8B, is no more accurate on AIME, because W4A4 costs it most of its lead over 8B.

1 Introduction

Reasoning models answer a competition problem by writing thousands to tens of thousands of tokens first (DeepSeek-AI et al., 2025; Yang et al., 2025), so their serving cost is decode cost, paid per generated token and per byte of KV cache held while the trace grows. Blackwell GPUs add native FP4 tensor cores for NVFP4 (E2M1 values in blocks of 16 with an E4M3 scale per block and an FP32 scale per tensor) (NVIDIA et al., 2025; Rouhani et al., 2023), which promise several times the BF16 throughput on one card, or a larger model in the same memory (Egiazarian et al., 2025). Three questions are open: whether a reasoning model keeps its accuracy under FP4, which part of the error matters, and what each fix costs in serving. Low-bit post-training quantization (PTQ) degrades reasoning accuracy (Liu et al., 2025a; Zhang et al., 2025a; Zheng et al., 2025) and lengthens chains of thought, which can cancel the per-token speedup (Lian et al., 2026; Lotfi et al., 2026; Alimaskina et al., 2026), while serving stacks compress the KV cache to FP8 or NVFP4 and report it nearly lossless on large models (vLLM Team, 2026; LMSYS Org, 2026).

Answering these questions runs into a measurement problem first. Methods are ranked by one checkpoint each on a benchmark of 30 problems per year with a handful of samples per problem. This resolves differences of about ±4.7 AIME points even when three years are pooled and comparisons are paired, and single runs of one checkpoint, or of two calibration draws, differ by as much as the published NVFP4 recipes differ from one another (Table 1). We therefore build the study around controls that separate one factor at a time, and report which conclusions hold up under them (Figure 1). The contribution is a measurement methodology and the deployment decisions it supports:

- A measurement protocol for quantized reasoning models (Section 3): paired per-problem tests pooled over AIME years with a power analysis for the number of samples and problems, repeated calibration draws, accuracy at every token budget computed exactly from one run, continuation of truncated samples past the cap, bit-identical-weight controls, and Holm correction over the paper's claims of a difference, released with

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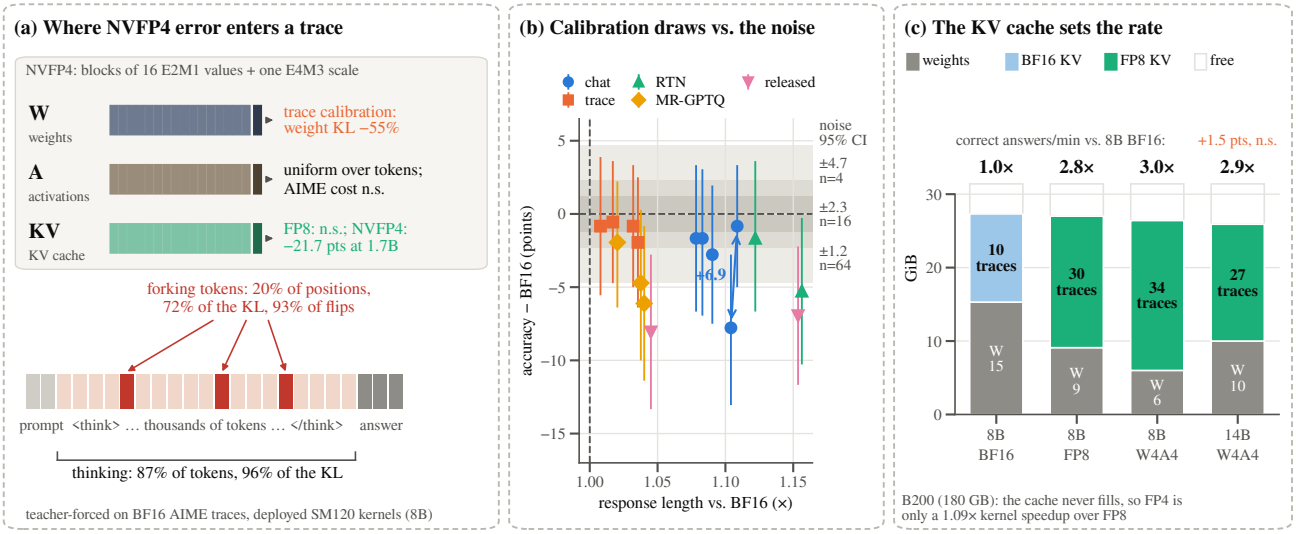


Figure 1. The study at a glance. (a) NVFP4 quantizes each layer's weights, activations and, optionally, its KV cache. Teacher-forced on BF16 AIME traces with the deployed SM120 kernels, the divergence of 8B W4A4 concentrates in the thinking segment and at its forking tokens; calibrating GPTQ on reasoning traces removes most of the weight part, and an FP8 cache adds little error, while an NVFP4 cache at 1.7B adds more than W4A4 itself (Section 4). (b) Every 8B W4A4 checkpoint against BF16 on AIME 2024–2026 (90 problems, 4 samples each, paired 95% intervals): chat-calibrated GPTQ draws are longer than every trace-calibrated one, and the arrow joins two runs of one of them that differ only in the sampling seed (6.9 points); the shaded bands are the half-width a paired comparison reaches with 4, 16 and 64 samples per problem, a lower bound for effects that vary between problems (Sections 3 and 5). (c) Memory of the 32 GB RTX 5090 per configuration, the traces its KV cache holds, and the measured rate of correct answers relative to 8B BF16: the FP8 cache gives most of the rate, FP4 weights add little at 8B, and the Qwen3-14B they fit at about the rate of 8B in FP8 is no more accurate on AIME (+1.5 points, Holm-adjusted $p = 0.304$), as W4A4 costs it most of its lead over 8B (Section 6).

a planner that turns the noise model into sample sizes and the capacity model into serving rates.

- An anatomy of the NVFP4 error on deployed SM120 kernels (Section 4, Figure 1a). Calibrating GPTQ on reasoning traces cuts the weight divergence by 55% (8B) and 60% (1.7B), after which activations carry most of it, so which component dominates depends on the calibration. Activation error is uniform over tokens, so token-adaptive precision has only proportional returns. Per-token error ranks checkpoints by end-to-end accuracy and length (Spearman −0.72 and 0.86 at 8B) and flags the NVFP4 KV collapse of the 1.7B model in advance.

- End-to-end comparisons of calibration data and published recipes (Section 5, Figure 1b). Calibrating on reasoning traces, the model's own or public ones, removes most of the extra length of chat calibration in every draw on both GPUs, on Qwen3 (a Llama-based distilled model shows none to remove); its accuracy gain over chat (+2.4 pooled points on the RTX 5090, +1.8 on the B200) is not significant. At 8B, trace GPTQ leads all three published NVFP4 recipes (MR-GPTQ, QAT, ModelOpt) on both GPUs, significantly over the released QAT and ModelOpt checkpoints and over MR-GPTQ on the RTX 5090 alone, and MR-GPTQ matches it when calibrated on the same traces; at 14B trace GPTQ and MR-GPTQ are equal (+0.3 [−2.0, +2.6], 16 samples). Even so, 8B W4A4 trace GPTQ is −2.6 [−4.8, −0.4] points from BF16 over both GPUs' samples, where FP8 with an FP8 cache is +0.2. On trace-calibrated weights activation error is equivalent to none within 2.4 points, so the FP4 cost is the weights' and full-row NVFP4 residuals, which bring per-token error down to the weight-only level, add decode time without a measurable gain.

- A capacity model of long-trace serving (Section 6, Figure 1c). On a 32 GB card the number of traces the KV cache holds sets the serving rate. Built from memory breakdowns and steady-state measurements at fixed concurrency, the model predicts each format's correct answers per minute within 5% on average, and within 7% for 14B when fitted on 8B alone. On a cache-bound card an FP8 cache is necessary, and it is equivalent to BF16 KV within 2.8 points; FP4 weights add only 1.09× over FP8 at 8B (1.05× at both GPUs' accuracies); and Qwen3-14B, which they fit at 1.04× the rate of 8B in FP8, is no more accurate on AIME, since W4A4 costs it −2.6 points of its +4.0-point lead.

2 Background and Related Work

FP4 formats and Blackwell kernels. NVFP4 stores E2M1 values (magnitudes $\{0, 0.5, 1, 1.5, 2, 3, 4, 6\}$) in blocks of 16 with an E4M3 block scale and an FP32 per-tensor scale; MXFP4 uses blocks of 32 with a power-of-two scale (Rouhani et al., 2023). Blackwell tensor cores multiply such operands natively, which has been used for FP4 training (NVIDIA et al., 2025; Castro et al., 2025) and inference. Egiazarian et al. (2025) show that NVFP4's small blocks neutralize rotation-based outlier handling and propose MR-GPTQ, a format-aware GPTQ variant; Cook et al. (2025) choose between two block scales. On SM120 a serving stack can run W4A4 GEMMs (CUTLASS, through FlashInfer (Ye et al., 2025)), weight-only W4A16 (Marlin (Frantar et al., 2025)) and FP8 W8A8 (Micikevicius et al., 2022); we measure all three end to end in vLLM (Kwon et al., 2023).

Post-training quantization of LLMs. GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) quantize weights from a small calibration set; SmoothQuant (Xiao et al., 2023) and rotations (Ashkboos et al., 2024; Liu et al., 2025b) move activation outliers out of the way of 4-bit activations; mixed-precision schemes keep outlier channels or a low-rank branch in higher precision (Dettmers et al., 2022; Zhao et al., 2024; Saxena et al., 2025; Li et al., 2025; Chen et al., 2024a; Lin et al., 2025). Calibration data matters (Williams & Aletras, 2024), more so for reasoning models, where its domain changes GPTQ results (Liu et al., 2025a). Closest to our calibration result, AYOT (Wang et al., 2026b) uses the model's own reasoning traces as calibration context for ternary PTQ. We study NVFP4 on deployed kernels, find public traces from another model as good as the model's own, and measure the effect against repeated calibration draws, which separates the part that survives (length) from the part that does not (accuracy at 8B). Token-adaptive precision routes tokens to different weight bit widths (Wang et al., 2026a); we ask the analogous question for NVFP4 activations.

Quantized reasoning models. Benchmarks of quantized reasoning models report accuracy loss that grows with task difficulty and lower bit widths (Liu et al., 2025a; Zhang et al., 2025a; Zheng et al., 2025; Mekala et al., 2025). Quantized reasoning models write longer chains of thought (Lian et al., 2026), concentrate their divergence at high-entropy positions and over-produce hesitation markers (Lotfi et al., 2026), and at 2 bits fall into loops (Alimaskina et al., 2026). Remedies retrain the model: ReQAT (Lee et al., 2026) with FP4 quantization-aware training, and Right On Q (Zheng et al., 2026) with on-policy distillation on the quantized student's own trajectories at 2–3 bits; or they penalize hesitation at decoding time (Lotfi et al., 2026). We stay with PTQ and vary its calibration data, which needs no training, and compare against a released QAT checkpoint. Because accuracy depends on the token budget (Muennighoff et al., 2025; Snell et al., 2025; Chen et al., 2024b), we report it at every budget and past the cap. That a few high-entropy "forking" tokens decide a trajectory (Bigelow et al., 2024; Wang et al., 2025) motivates our position-resolved analysis, and we report flips alongside accuracy (Dutta et al., 2024).

KV-cache quantization. KIVI (Liu et al., 2024) and KVQuant (Hooper et al., 2024) quantize keys per channel and values per token at 2–4 bits; recent work adapts KV precision to queries or to the structure of reasoning traces (Zhang et al., 2025b; Ramachandran et al., 2025). In deployed stacks the KV cache is FP8 or NVFP4 with a per-tensor scale that defaults to 1.0; on models of 27B parameters and more this is reported as nearly lossless on AIME (vLLM Team, 2026; LMSYS Org, 2026). We measure the same formats on smaller models and long traces, by position and with calibrated scales.

3 Setup and Measurement Protocol

Models and serving. We study Qwen3-1.7B, 8B and 14B (Yang et al., 2025) in thinking mode with vLLM 0.32, FlashInfer 0.7 and CUDA 13.0, on one RTX 5090 (32 GB, SM120) and, for replication, one B200 (180 GB, SM100). BF16 is the reference at 1.7B and 8B; on the 5090, BF16 14B weights (29.5 GB) leave no room for a KV cache, so Qwen's FP8 checkpoint with an FP8 cache is the 14B reference there and BF16 14B runs on the B200. FP8 W8A8 quantizes weights offline and activations dynamically. Our NVFP4 checkpoints are written by LLM Compressor (Red Hat AI and vLLM Project, 2024); W4A4 runs on the CUTLASS FP4 GEMM and W4A16 on Marlin (Table 8). Activation global scales are static per input site; block scales are computed per token at run time; the `lm_head` stays in BF16.

Weights. RTN rounds every weight to the nearest E2M1 value under its block scale. Chat GPTQ calibrates GPTQ on 256 UltraChat conversations of up to 2048 tokens; draws 2–4 take the next 256 conversations each. Trace GPTQ calibrates it on reasoning traces the model writes itself in thinking mode for problems from the MATH training split (Hendrycks et al., 2021), disjoint from every evaluation set, cut at 4096 tokens (128 traces at 8B and 14B, the token budget of chat calibration; 256 at 1.7B); a second trace set uses other problems and seeds, and R1 GPTQ uses public DeepSeek-R1 traces from OpenR1-Math in the same budget. The W4A4 and W4A16 trace checkpoints hold bit-identical weights (checked tensor by tensor), so comparing them isolates activation quantization. The chat checkpoints do not: LLM Compressor shuffles chat samples in an unseeded order, so each chat checkpoint is an independent GPTQ draw.

MR-GPTQ (Egiazarian et al., 2025) is quantized with the authors' code and published NVFP4 recipe (Hadamard blocks of 16, MSE-searched scales, activation order, FineWeb-Edu calibration at our 256×2048 budget), with two FineWeb draws and one calibration on our traces; ISTA's QAT checkpoints (1.7B, 8B) and NVIDIA's ModelOpt checkpoint (8B) are used as released. FP-Quant's fused activation kernel is built for SM100 only, so on SM120 MR-GPTQ and QAT run through our port of the same arithmetic (Appendix D), equal in accuracy but with an unfused rotation: their SM120 throughput is a lower bound, and on the B200 they run natively.

Residual activations, W4A(4+4), and the KV cache. To restore activation precision on FP4 tensor cores, each activation row a is quantized twice, $q_1 = Q(a)$ and $q_2 = Q(a - \mathrm{DQ}(q_1))$, and one NVFP4 GEMM runs over the stacked rows $[q_1; q_2]$; a fused Triton kernel produces both. On a fraction ρ of rows chosen by a score this is token-adaptive precision (Section 4.3); on every row it is W4A(4+4). The KV cache is BF16, FP8 E4M3 or NVFP4 (FlashInfer; queries stay in BF16 on SM120 and are quantized for FP8 attention on SM100) with a per-layer decode scale that is 1.0 unless calibrated (Appendix H); every quantized-KV accuracy is compared with BF16 KV on the same backend.

Evaluation. We evaluate on AIME 2024, 2025 and 2026 (Balunović et al., 2025): 90 problems on which Qwen3 writes traces of 14–19k tokens on average, with Qwen3's recommended sampling (temperature 0.6, top-p 0.95, top-k 20), 4 samples per problem (16 for the capacity comparison) and a cap of 30,720 response tokens, graded with Math-Verify (Kydlíček, 2025); GPQA-Diamond (Rein et al., 2024) and IFEval (Zhou et al., 2023) check other domains. Because a sample that finishes within a budget B is generated identically under any cap $\geq B$, accuracy at every budget up to the cap follows exactly from one run; samples that hit the cap are continued from their own tokens to 40,000 tokens, Qwen3's position limit. Teacher-forced replays of BF16 traces (32 MATH-500 traces, 150–170k positions per model; the AIME traces of each year, up to 32k tokens, for the KV cache) through a simulator that rounds scales and values as the kernels do (Appendix C) give $\mathrm{KL}(p_{\mathrm{BF16}} \,\|\, q)$ and top-1 flips per position; W4 quantizes only weights, A4 only activations. Every checkpoint is also screened on its own serving kernels by the excess NLL of BF16 AIME traces (Appendix E).

Statistics, and what the noise allows. Every comparison is paired: the difference of per-problem accuracies is resampled over problems (10,000 bootstrap draws) with the AIME years pooled, and its p-value comes from random sign flips of the per-problem differences. Table 1 measures the noise.

Table 1. What moves an 8B AIME result: paired differences (pooled 2024–2026, 4 samples per problem) between runs that differ only in the factor named, and the 95% half-width of a paired comparison of two configurations by samples per problem n and problems P (from the variance decomposition of Section 3). SD across draws: beyond sampling, with its 95% upper bound. The half-widths assume no between-problem variation of the true difference, as measured for BF16 against trace GPTQ, and are lower bounds where an effect varies between problems.

Factor (same everything else)	Δ AIME [95% CI]
Sampling seed, one checkpoint	−0.3 [−4.4, +3.9]
the same for RTN	−3.6 [−7.5, +0.0]
Attention kernel (FA2 vs. FlashInfer)	−3.3 [−7.8, +0.8]
the same for chat draw 1	+6.9 [+2.5, +11.7]
Chat calibration draw (1 vs. 2)	+6.1 [+1.7, +10.6]
with both seeds of draw 1	+2.6 [−1.2, +6.4]
the same two on the B200	+1.1 [−3.6, +6.1]
Trace calibration draw (1 vs. 2)	−1.4 [−5.8, +2.8]
the same two on the B200	+1.9 [−3.1, +6.7]
RTN (two seeds) vs. ModelOpt	+3.5 [+0.1, +6.8]
the same on the B200, $n = 16$	−1.5 [−4.0, +0.8]
SD across draws: chat (4)	2.5 (≤8.4)
trace (3)	0.0 (≤2.8)
on the B200: chat (4) / trace (3)	0.0 (≤3.3) / 0.0 (≤5.6)
Half-width, $n = 4$: $P = 30$ / 90 / 180	8.1 / 4.7 / 3.3
Half-width, $n = 16$: $P = 30$ / 90 / 180	4.1 / 2.3 / 1.7
Half-width, $n = 64$: $P = 30$ / 90 / 180	2.0 / 1.2 / 0.8

For two configurations over P problems with n samples each, the variance of the mean difference is $(\sigma_\Delta^2 + \bar{v}/n)/P$, where $\bar{v}$ is the mean binomial variance of a sample and σ_Δ^2 the between-problem variance of the true per-problem difference. For BF16 against W4A4 trace GPTQ the observed spread is explained by sampling alone ($\hat{\sigma}_\Delta = 0.0$), so the half-width shrinks as $1/\sqrt{nP}$: four samples on 90 problems resolve ±4.7 points and a single year ±8.1. Whether calibration draws add a second source, one that does not shrink with n, is less clear than a single GPU suggests. On the RTX 5090 two chat-calibrated GPTQ checkpoints differ by +6.1 [+1.7, +10.6] points with equal per-token error (ΔNLL 0.0428 vs. 0.0427), but a second seed moves the first by +6.9 [+2.5, +11.7], after which the two are +2.6 [−1.2, +6.4] apart, so the gap came from one low run, and a third and a fourth draw score 65.3% and 64.2%, near the second; on the B200 the same two differ by +1.1, and its 4 chat, 3 trace and 2 MR-GPTQ draws spread no more than sampling allows (between-draw SD at most 3.3 points for chat and 5.6 for trace, 95% upper bounds; two MR-GPTQ draws bound it only loosely). Such tails are somewhat more common than the intervals imply: over 21 pairs of runs that differ only in the sampling seed, the paired 95% intervals miss 1 of the differences, yet the differences spread 1.23 times as widely as the intervals imply ($\sum z^2 = 31.8$, $p = 0.06$), so a single-run comparison deserves an interval about that much wider (RTN: Section 5.5). A method is therefore

represented by several draws where we can make them, the decisive comparisons are repeated with 16 samples on the B200, and published checkpoints, single draws by construction, are read with that spread in mind. The paper's claims of a difference form two families, Holm-corrected separately: 9 comparisons on the RTX 5090 and 6 with 16 samples on the B200 (Table 6). Each holds every difference the text claims; they were fixed as the text was written, not pre-registered. Claims of no difference are stated with their intervals and, where a recommendation rests on one, with the margin within which a two-one-sided test (TOST, 5% per side) shows equivalence.

4 Where the FP4 Error Comes From

All numbers in this section are teacher-forced (Section 3) on 32 BF16 MATH-500 traces per model, unless stated otherwise; KL is in nats per response token.

4.1 Weights or activations

Figure 2 decomposes the W4A4 divergence. With chat-calibrated GPTQ, 8B weights alone give KL 0.018 and activations alone 0.018, and the two parts add up almost exactly (W4A4: 0.036). Activations carry 51% of the KL but only 31% of the top-1 flips: weight error moves the argmax more often per unit of KL. The 1.7B model diverges three times as much at every component (W4 0.059, A4 0.048) with a similar split (49% / 29%). GPTQ on chat data reduces the weight KL of RTN by only 32% (8B) and 23% (1.7B).

The calibration data moves the split much further than the algorithm does. GPTQ calibrated on the model's own reasoning traces cuts the weight KL by 55% at 8B (to 0.0085) and by 60% at 1.7B (to 0.024), while the activation part is unchanged. Activations then carry 70% of the 8B KL and 47% of its flips (1.7B: 68% / 43%). Whether an FP4 deployment should spend effort on weights or on activations therefore has no format-level answer: with chat-calibrated or RTN weights it is the weights, with trace-calibrated weights it is the activations. Up-weighting high-entropy positions in the GPTQ Hessian does not help (Appendix J).

The share does not predict the end-to-end cost. On trace-calibrated weights, W4A4 and W4A16 differ only in activation quantization (bit-identical weights), and at neither size does the difference register on AIME: −0.3 [−2.8, +2.3] points at 8B over both GPUs' 20 samples, equivalent to none within 2.4 (activation KL 0.019), and −0.8 [−5.0, +3.3] at 1.7B (0.048), although activations carry most of the remaining per-token KL. Divergences of this size do not decide AIME problems; the NVFP4 KV cache at 1.7B, at four times the activation KL, does (Sections 4.4 and 5.4).

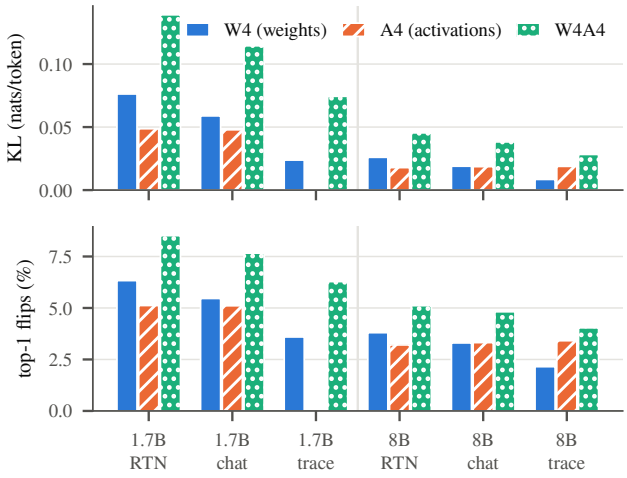


Figure 2. Teacher-forced KL (top) and top-1 flips (bottom) of weights only (W4), activations only (A4) and both (W4A4), for RTN, chat-calibrated and trace-calibrated GPTQ weights at both sizes. Trace calibration removes most of the weight part; the activation part does not depend on the weights (A4 was not rerun for 1.7B trace).

4.2 Where in the trace the divergence lives

The divergence concentrates where the BF16 model is uncertain (Figure 8). Positions with BF16 next-token entropy of at least 0.5 nats are 20% of the 8B response positions and carry 72% of its W4A4 KL mass and 93% of its flips (1.7B: 20%, 70%, 84%). The thinking segment holds 87% of the positions and 96% of the KL. These are the forking positions at which a sampled trajectory branches (Bigelow et al., 2024; Wang et al., 2025), in line with the high-KL, high-entropy positions of Lotfi et al. (2026). Near-deterministic positions (entropy below 0.01 nats, 59% of positions) carry 2% of the KL and <1% of the flips; teacher forcing does not see errors that would cascade after a resampled token there, the mechanism Lee et al. (2026) target with quantization-aware training. Along long AIME traces the W4A4 error per token falls rather than grows (Section 4.4).

4.3 Token-adaptive activation precision

If a few tokens carried most of the activation error, giving only those rows a residual (Section 3) would recover most of the W4A4-to-W4 gap at a fraction of the cost. It does not (Figure 9 in Appendix F). Ranking rows by the norm of their own residual or by its effect on the layer output recovers 12% and 15% of the activation part at 10% of rows per site, no more than random rows (12%): NVFP4's per-block scaling leaves every token with about the same relative error. Only oracles do better: the BF16 entropy of the predicted token, known only after the forward pass, recovers 27% (the previous step's entropy, its online counterpart, 13%), and

a two-pass oracle that ranks positions by their measured W4A4 KL 34%. Residuals on every row recover all of it: W4A(4+4) has KL 0.018, the weight-only level 0.018. The picture is the same at 1.7B and on trace-calibrated weights, where activations carry most of the divergence (Appendix G). Since a data-dependent row selection also costs a gather and an uneven GEMM shape, the practical choices are all rows or none; Section 6.1 measures the cost of all rows.

4.4 The KV cache, by position

We teacher-force the 30 BF16 AIME traces of each year and model (up to 32k response tokens) with a quantized cache. Two regimes appear. At 8B the cache's per-token error does not grow with context: over the 10 AIME 2024 traces that reach 16k tokens, FP8 KV has 0.97× [0.70, 1.22] the KL beyond 16k that it has in the first 8k, NVFP4 KV 0.83× [0.56, 1.09]. Per token, an 8B NVFP4 cache costs nearly as much as W4A4 on every linear layer (KL 0.023 vs. 0.027, trace-GPTQ weights), an FP8 cache about a fifth of it (0.0047), and on W4A4 weights the errors roughly add (0.032 and 0.055). At 1.7B the cache dominates: NVFP4 KV has KL 0.193 against 0.078 for W4A4 (flips 9.9 vs. 6.8%), more than additive on W4A4 weights (0.326), and the FP8 error grows along nearly every long trace (11 of 11; 1.90× [1.61, 2.22] beyond 16k; the 2025 and 2026 traces agree, Appendix H). The error of the quantized linears falls along the trace at both sizes (W4A4: 0.64× at 8B, 0.85× at 1.7B), so the cache's share of the total grows with length.

With the default decode scale $s = 1$ many layers store values below E4M3's normal range: at 8B, 6.40% of the NVFP4 value blocks have a subnormal block scale and 7.95% of FP8 values are subnormal (Appendix H). Calibrated per-layer scales remove this underflow but change the error little (8B: FP8 0.0044, NVFP4 0.022; 1.7B: NVFP4 0.185, FP8 growth 1.63×), so the cache error is rounding on the normal grid.

4.5 What per-token error predicts end to end

Across the W4A4 checkpoints with end-to-end runs (23 at both sizes; the 1.7B QAT checkpoint, which no longer thinks, is excluded), the deployed ΔNLL ranks the pooled AIME change from BF16 (Spearman −0.81 [−0.94, −0.54]; 8B alone −0.72 [−0.94, −0.12]) and, at 8B, the length change (0.86 [0.50, 0.98]) (Figure 3). It rates checkpoints a few thousandths of a nat apart as equal, and their AIME gaps do not reproduce either: the two chat draws (0.0428 and 0.0427) differ by +2.6 points on the RTX 5090 over both seeds of the first and +1.1 on the B200, RTN and ModelOpt (0.0558 and 0.0548) by +3.5 and −1.5. Per-token error orders families of checkpoints and flags gross failures, like the 1.7B NVFP4 cache (Section 5.4); it does not replace repeated end-to-end runs, and a threshold for the KV decision rests on three models only.

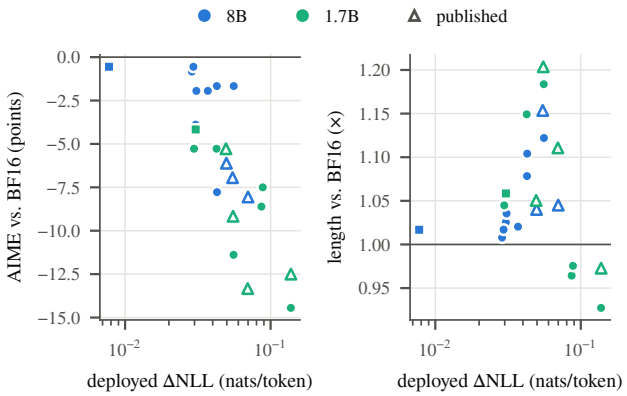


Figure 3. Deployed per-token error (excess NLL on BF16 AIME traces) against the pooled AIME change (left) and response length (right) relative to BF16, for every W4A4 checkpoint with end-to-end runs. Squares: W4A(4+4); triangles: published checkpoints.

5 End-to-End Accuracy and Length

5.1 Calibration data

Table 2 gives every configuration and Figure 1b and Figure 6 every 8B W4A4 checkpoint. FP8 W8A8 is within noise of BF16 (−2.5). Against BF16, chat GPTQ pooled over its draws is −3.5 [−7.7, +0.6] at 1.09× the BF16 length, and trace GPTQ pooled over its three draws −1.1 [−4.9, +2.6] at 1.02×; trace over chat is +2.4 [−0.2, +5.1] points at 0.94× the tokens ($p = 0.094$, Holm-adjusted 0.188). On both GPUs every chat draw writes longer traces than every trace draw (Figure 1b), while the accuracy difference is not significant on either. On the B200, pooled over four chat and three trace draws, trace over chat is +1.8 [−1.1, +4.6] points at 0.94× the tokens, and the draws of neither source spread more than sampling allows (Table 1). What FP4 costs against BF16 needs more than four samples: with 16 per problem on the B200, trace GPTQ is −3.1 [−5.6, −0.8] from BF16 and FP8 with FP8 KV −0.3 [−2.2, +1.5], and over both GPUs' samples trace GPTQ is −2.6 [−4.8, −0.4] (every recipe at 16 samples: Table 9). The cost is the weights': W4A16 with the same weights is −2.6 [−5.0, −0.3] from BF16 with 16 samples, and activations add none measurable (Section 4.1). On the RTX 5090, RTN, which needs no calibration data for its weights, is within the noise of trace GPTQ over two sampling seeds (+2.4 [−0.9, +5.7]) at 1.14× the BF16 length; on the B200 with 16 samples per problem it trails trace GPTQ by +6.7 [+3.8, +9.6] (Section 5.5). Public R1 traces calibrate as well as the model's own (−0.3 at 8B, −1.1 at 1.7B, equal per-token error), so no generation step is needed.

Table 2. End-to-end AIME accuracy (%) on the RTX 5090, pooled over 2024–2026 (90 problems, 4 samples each). Δ: paired difference to BF16 of the same size with a 95% bootstrap interval over problems; tok: total response tokens relative to BF16. Truncated samples count as wrong at the 30,720-token cap (40k: Section 5.2). Pooled rows pool every draw of a source (chat: 4 draws; trace: two self-generated sets and R1; MR-GPTQ: FineWeb draws). Quantized-KV runs use FlashInfer; at 1.7B they cover AIME 2024 only (in the text). Per year: Table 10. –: not run.

	Qwen3-1.7B			Qwen3-8B		
Configuration	AIME	Δ [95% CI]	tok	AIME	Δ [95% CI]	tok
BF16	40.3	–	–	66.9	–	–
FP8 W8A8	–	–	–	64.4	−2.5 [−7.5, +2.5]	1.02
W4A16 chat GPTQ	34.2	−6.1 [−11.4, −0.8]	0.99	64.7	−2.2 [−8.1, +3.3]	1.08
W4A16 trace GPTQ	33.6	−6.7 [−11.9, −1.7]	1.04	65.6	−1.4 [−5.8, +3.1]	1.01
W4A4 RTN	–	–	–	65.3	−1.7 [−6.9, +3.3]	1.12
W4A4 chat GPTQ, draw 1	25.8	−14.4 [−20.3, −8.9]	0.93	59.2	−7.8 [−13.1, −2.8]	1.10
W4A4 chat GPTQ, all draws	–	–	–	63.5	−3.5 [−7.7, +0.6]	1.09
W4A4 trace GPTQ, draw 1	32.8	−7.5 [−12.5, −2.2]	0.98	66.4	−0.6 [−4.7, +3.6]	1.02
W4A4 R1 GPTQ	31.7	−8.6 [−13.9, −3.3]	0.96	66.1	−0.8 [−5.6, +3.9]	1.01
W4A4 trace GPTQ, all draws	–	–	–	65.8	−1.1 [−4.9, +2.6]	1.02
W4A(4+4) trace GPTQ	36.1	−4.2 [−9.2, +0.6]	1.06	66.4	−0.6 [−5.0, +3.9]	1.02
MR-GPTQ (published recipe)	27.8	−12.5 [−17.8, −7.5]	0.97	60.8	−6.1 [−11.7, −0.8]	1.04
MR-GPTQ, all FineWeb draws	–	–	–	61.5	−5.4 [−10.0, −1.0]	1.04
MR-GPTQ on our traces	–	–	–	65.0	−1.9 [−6.4, +2.2]	1.02
ISTA QAT (released)	2.5	−37.8 [−45.6, −30.0]	0.41	58.9	−8.1 [−13.6, −2.8]	1.05
NVIDIA ModelOpt (released)	–	–	–	60.0	−6.9 [−11.9, −2.2]	1.15
FP8 W8A8 + FP8 KV	–	–	–	67.2	+0.3 [−4.4, +4.4]	1.01
W4A4 trace + FP8 KV	–	–	–	65.0	−1.9 [−6.4, +2.2]	1.03
W4A4 trace + NVFP4 KV	–	–	–	62.5	−4.4 [−9.7, +0.6]	1.08
W4A(4+4) trace + FP8 KV	–	–	–	67.5	+0.6 [−3.6, +4.4]	1.01

At 1.7B quantization costs more. Weight-only W4A16 loses −6.1 [−11.4, −0.8] with chat and −6.7 [−11.9, −1.7] with trace calibration, although the latter has 60% less weight KL. Chat W4A4 loses −14.4 [−20.3, −8.9]; trace W4A4 recovers +6.9 [+1.9, +12.2] of it (Holm-adjusted $p = 0.054$), and on its bit-identical weights the activations account for −0.8.

Outside competition math. Calibrating on mathematical traces costs no accuracy elsewhere. On the B200, on GPQA-Diamond (thinking, 198 questions, 4 samples each), trace GPTQ is −3.5 [−6.5, −0.6] points from BF16 and chat GPTQ −5.4 [−8.1, −2.7] (two draws each), and trace over chat is +1.8 [−0.7, +4.4] at 0.94× the tokens, so FP4 costs about as much on GPQA as on AIME and trace calibration keeps its shorter traces. IFEval and MMLU-Pro, with thinking off, do not separate the calibration sources either (Appendix K).

What the calibration has to see. Three ablations at trace calibration's token budget, two draws each, separate its ingredients by their effect on length, the signal that clears the noise (8B on the B200, paired against every chat and every trace draw there; traces write 0.94× chat's tokens; Table 9). Chat packed into 4096-token sequences writes as long as ordinary chat calibration (1.00× [0.98, 1.03]), so sequence length does not explain the effect. The model's own thinking traces on programming problems, thinking without the domain, write as long as math traces (1.01× [0.99, 1.04]); MATH problems with human-written solutions, the domain without the model's thinking, fall between (1.02× [1.00, 1.05] that of math traces). GPTQ needs the model's thinking-mode text, from any domain; accuracy does not separate the sources. On DeepSeek-R1-Distill-Llama-8B, chat-calibrated W4A4 GPTQ is −0.3 [−6.1, +5.8] points from BF16 at 1.05× [0.99, 1.11] its tokens (Qwen3-8B: 1.12×), and public R1 traces change it by +3.1 [−1.7, +7.5] points and 1.04× in length: chat calibration adds no measurable length there, so its length effect is not universal.

5.2 Accuracy at every budget

Figure 4 plots accuracy against the response budget. Chat-calibrated 8B W4A16 and W4A4 (draw 1) trail BF16 at every budget (W4A4: −5.3 at 8k, −4.2 at 16k, −7.8 at the cap), while trace GPTQ stays within +1.7 at 16k and −0.6 at the cap. Continuing every truncated sample to 40k tokens leaves the gaps where they were (chat W4A4 −5.8, trace −2.2), so the cap decides only whether a failing trajectory counts as truncated or as wrong. Continued samples rarely end correct (99 of 962 over the quantized 8B configurations, 1 of 352 at 1.7B), and many of those still running at 40k repeat themselves (Table 11; the count is insensitive to the loop threshold). The extra tokens of the FP4 models go to trajectories that fail.

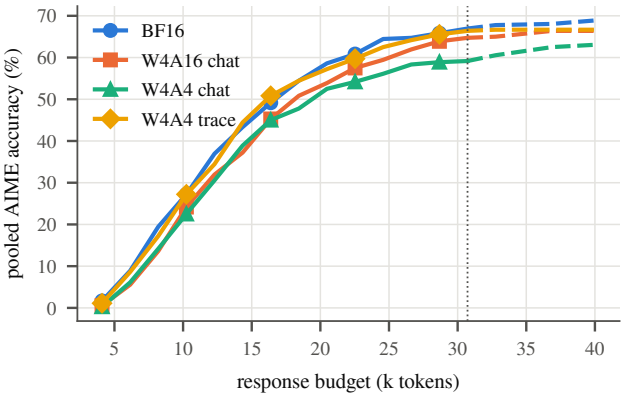


Figure 4. Pooled AIME accuracy vs. response budget, Qwen3-8B: exact from one run up to the 30,720-token cap (dotted), dashed beyond it from the continuation of truncated samples. Paired differences to BF16 are in the text.

5.3 Residual activation rows

W4A(4+4) on trace weights brings 1.7B W4A4 to −4.2 [−9.2, +0.6] of BF16, +3.3 [−0.8, +7.5] over W4A4-trace at 1.09× its tokens; at 8B the rows change +0.0 [−4.2, +4.2]. The bit-identical W4A16 control predicts this: on trace-calibrated weights the activation part costs no measurable accuracy at either size (Section 4.1), so the rows have little to recover, while they add decode time (Section 6.1).

5.4 KV compression on long traces

Against a BF16-KV run on the same FlashInfer backend, an FP8 cache changes 8B W4A4-trace accuracy by +1.9 [−2.2, +6.1], a calibrated FP8 cache by +1.7 [−3.3, +6.7] and an NVFP4 cache by −0.6 [−5.0, +3.9] at 1.05× the tokens. The backend swap itself, FlashAttention to FlashInfer with BF16 KV, moves accuracy by −3.3 [−7.8, +0.8], within noise, and teacher-forced on identical weights the two kernels differ by at most 0.0001 nats/token (95% bound), against 0.0049 for FP8 weights. At 1.7B, on AIME 2024, FP8 KV changes accuracy by +1.7 [−7.5, +10.8] although its per-token error grows along the trace (Section 4.4), and NVFP4 KV costs −21.7 [−32.5, −11.7] (problems won 1:15; Holm-adjusted $p = 0.003$): the outputs stay fluent but their arithmetic and self-checks fail. The per-token divergence gives this ordering in advance: an NVFP4 cache adds 0.023 nats/token at 8B, less than W4A4 itself (0.027), and 0.193 at 1.7B, more than twice W4A4's 0.078. At 14B, NVFP4 KV changes W4A4-trace accuracy by −3.9 [−8.9, +0.8] against FP8 KV. FP8 KV has no measurable cost at 8B on either GPU (Section 6.5) or at 1.7B, within a wide one-year interval. The NVFP4 result does not transfer to SM100: there vLLM also quantizes queries to FP8 for an NVFP4 cache, and 8B W4A4 trace loses −5.9 [−8.8, −3.2] against BF16 KV on the B200 with 16 samples per problem (Section 6.5).

5.5 Against published NVFP4 recipes

Pooled over AIME at 8B, MR-GPTQ is −6.1 [−11.7, −0.8] from BF16 (all FineWeb draws: −5.4), ISTA's QAT checkpoint −8.1 [−13.6, −2.8] and NVIDIA's ModelOpt checkpoint −6.9 [−11.9, −2.2]. Each is at or below our RTN checkpoint (−3.5, two seeds). Trace GPTQ, pooled over its draws, leads them by +4.3 [+1.2, +7.4], +6.9 [+2.9, +11.4] and +5.8 [+2.3, +9.4] points; after Holm correction the leads over QAT ($p = 0.014$) and ModelOpt (0.013) remain, and the lead over both MR-GPTQ draws just does (0.047). With 16 samples per problem on the B200, where every method runs on its native kernel, trace GPTQ is +3.1 [+0.4, +5.9] from MR-GPTQ, +5.1 [+2.2, +8.1] from ModelOpt, +8.1 [+5.0, +11.3] from QAT and +6.7 [+3.8, +9.6] from RTN. Of these 6 comparisons, 3 survive Holm correction: the leads over ModelOpt ($p = 0.004$), QAT (0.000) and RTN (0.000); the leads over MR-GPTQ (0.095) and chat GPTQ (+2.3, 0.178) do not. RTN's place among the recipes rests on one run. Above every published recipe in its first four-sample run on the RTX 5090, it changes by −3.6 [−7.5, +0.0] at a second seed there and is −3.6 [−6.5, −0.9] from MR-GPTQ and −1.5 [−4.0, +0.8] from ModelOpt with 16 samples on the B200. Of the samples it finishes, that first run answers 81.9% correctly, against 77.1, 75.5 and 76.7% in RTN's three other runs, and against the RTX 5090's second seed alone the B200 is within the noise (−2.8 [−6.0, +0.3]). The rest of RTN's cross-GPU gap (−4.6 [−7.4, −1.9] against both seeds) is truncation: 22.9% of RTN's samples hit the cap on the B200 against 20.1% (Δ +2.8 [+0.0, +5.6]), and repetition loops are as common among them (13% on the RTX 5090, 13% on the B200). Calibrated on our traces, MR-GPTQ changes by +3.5 [−0.1, +6.9] and trace GPTQ is +0.8 [−2.5, +4.6] from it, so most of the lead is the calibration data; per token, the Hadamard rotation still adds error on the same data (Appendix E), as Egiazarian et al. (2025) report for short tasks on Qwen3-8B. At 1.7B, trace GPTQ leads MR-GPTQ by +5.0 [+1.1, +9.2]; at 14B the two are equal: on the B200 over all three years, trace GPTQ is +0.3 [−2.0, +2.6] from a second-draw MR-GPTQ checkpoint with 16 samples (the RTX 5090's first draw, AIME 2024–2025: −2.9 [−8.3, +2.5]). The released 1.7B QAT checkpoint no longer thinks: nearly every sample opens with an empty think block, on our port and on FP-Quant's SM100 kernel alike (2.5% and 3.3%).

6 Long-Trace Serving Is Bound by KV Capacity

6.1 Kernel speed

With 128 input and 128 output tokens per sequence (Figure 7, Appendix F), decoding Qwen3-8B at batch 1 is memory-bound and any 4-bit weight format helps (W4A16 4.40 ms per step, W4A4 4.62, FP8 6.35, BF16 9.98).

Table 3. Memory and serving of AIME runs on the RTX 5090 (pooled years). W, KV: GiB from vLLM's accounting; K: KV cache (k tokens); L: mean response (k tokens); C^*: concurrent traces the cache holds; c/min: measured; err: capacity-model prediction of c/min relative to the measurement.

Configuration	W	KV	K	L	C^*	c/min	err (%)
Qwen3-8B							
BF16	15.3	12.0	88	16.6	10	0.97	+4
FP8	9.1	17.8	129	17.0	15	1.38	−3
+ FP8 KV	9.1	17.9	260	16.8	30	2.72	−5
W4A16 trace	6.0	20.8	151	16.9	18	1.64	−3
W4A4 trace	6.0	20.8	152	16.9	18	1.69	−5
+ FP8 KV	6.0	20.4	297	17.1	34	2.95	−8
+ NVFP4 KV	6.0	20.4	529	17.9	58	3.22	+13
Qwen3-14B							
FP8	15.4	11.0	72	15.1	9	1.04	+7
+ FP8 KV	15.4	10.6	139	15.1	18	1.92	+7
W4A4 trace	10.0	16.3	107	15.6	13	1.42	+2
+ FP8 KV	10.0	15.9	209	15.4	27	2.82	−2

W4A16 dequantizes in the kernel and computes in BF16, so at batch 256 it is almost as slow as BF16 (23.90 vs. 26.33 ms), while W4A4 keeps its lead (11.40 ms; prefill of 256 prompts 483 ms against 2195 for BF16). W4A(4+4) takes 9.31 instead of 7.69 ms per decode step at batch 128 (GEMM overheads: Appendix F).

6.2 Memory, concurrency and a capacity model

For long traces these kernel differences matter less than memory. A reasoning trace holds its whole context in the KV cache for tens of thousands of steps, so on a 32 GB card memory sets the number of traces decoded at once. Table 3 breaks the card down per configuration from vLLM's own accounting: weights W, KV memory and the cache size K in tokens. A trace of final length L holds on average $p + L/2$ tokens (p prompt tokens), so the cache sustains $C^* = \min(N, K/(p + L/2))$ concurrent traces for N requests. When the cache is full, each decode step streams the weights and the whole cache, $M = W + \text{KV}$, which is nearly the same for every configuration (26–27 GiB), plus a small cost per sequence in the batch. The step time is therefore $\tau = M/B + g\,C^*$, the throughput $T = C^*/\tau$, and the rate of correct answers $60\,Ta/L$ per minute for accuracy a. One fit of B and g over the 8B and 14B configurations gives $B = 1292$ GB/s (72% of the RTX 5090's peak) and $g = 0.20$ ms, with the memory term 86% of the step, and predicts the measured correct answers per minute of all 11 configurations within 5% on average (at most 13%, for the NVFP4 cache, whose concurrency is highest). Fitted on 8B alone, it predicts the four 14B configurations within 7% (12% at most); with each configuration left out in turn, within 8% on average, except the NVFP4 cache, alone at its concurrency (33%).

The cache's precision therefore matters as much as the weights': FP8 KV doubles K and, with it, throughput, and is equivalent to a BF16 cache within 2.8 points (Section 6.5). Longer traces cost twice, in fewer concurrent traces and in more tokens per answer, so correct answers per minute scale as aK/L^2 near full occupancy, and the chat draws' 6% longer traces cost about 12% of the rate of trace calibration before any accuracy loss. Once the cache is compressed, FP4 weights help only through the memory they free for it: at 8B, FP8 weights already leave most of the card to the cache, so W4A4 with FP8 KV holds 34 traces against 30 for FP8 with FP8 KV, and the best 8B FP4 configuration answers 1.09× as many problems per minute, at an accuracy +0.3 [−3.9, +4.7] from FP8's in this card's four-sample runs. Over both GPUs' samples, W4A4 trace with FP8 KV is −3.9 [−5.7, −2.2] points behind FP8 with FP8 KV, which brings its rate gain from 1.08× to 1.05×.

6.3 Steady state at fixed concurrency

Whole-run throughput mixes a full-batch phase with a tail of a few long traces. Figure 5 instead holds C requests in flight, each with 256 prompt and 8192 generated tokens. On the RTX 5090 throughput grows with C while the cache holds every request and flattens once vLLM has to preempt and recompute sequences. At $C = 64$ every configuration preempts, and the throughput ratios between configurations equal their cache-size ratios within 6%: FP8 with FP8 KV over BF16 2.85× (cache 3.01×), W4A4 over FP8 weights, both with FP8 KV, 1.16× (1.17×), and the same at 14B 1.54× (1.50×). The GEMM speedup of FP4 shows only at $C = 8$, where the cache is not full (1.23×). These are token throughputs; the correct-answer rates of Section 6.2 also carry each configuration's length and accuracy. On the B200 the cache of every configuration with FP8 KV holds 128 such requests, so throughput keeps growing and FP4 weights add 1.09× over FP8 at 8B and 1.16× at 14B through faster GEMMs.

6.4 A larger model at the same rate

By the capacity model, FP4 weights matter on this card where FP8 weights would crowd out the cache. Qwen3-14B in FP8 (15.4 GiB) leaves 139k tokens of FP8 cache; in NVFP4 (10.0 GiB) it leaves 209k, close to 8B in FP8 (260k). On AIME 2024–2026, 14B W4A4 trace GPTQ with an FP8 cache runs at 1.47× the rate of 14B FP8 with FP8 KV and 1.04× that of 8B FP8 with FP8 KV. With 16 samples per problem, its accuracy is +1.5 [−1.1, +4.1] relative to 8B FP8 + FP8 KV (Holm-adjusted $p = 0.304$), and +0.4 [−2.3, +3.1] with 16 samples on the B200 (+0.9 [−1.1, +3.1] with both GPUs' samples pooled): at the same rate, the larger model is no more accurate. On the B200, in FP8 with FP8 KV, Qwen3-14B leads 8B by +4.0 [+1.6, +6.6] (BF16: +3.0; Qwen's report: about three (Yang et al., 2025)), and W4A4

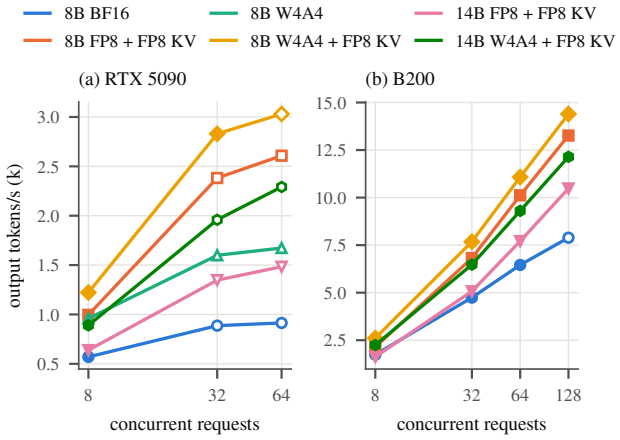


Figure 5. Steady-state output throughput at fixed concurrency (256 prompt, 8192 output tokens per request) on the RTX 5090 (a) and the B200 (b). Hollow markers: levels at which vLLM preempted sequences, i.e. the KV cache was full.

costs it −2.6 [−4.6, −0.6] against FP8 + FP8 KV over both GPUs (−1.2 on the RTX 5090, −3.6 on the B200), most of that lead. The capacity FP4 frees thus helps only a model whose lead over the smaller one exceeds its FP4 cost.

6.5 Datacenter Blackwell

On a B200 (SM100, 180 GB) with the same checkpoints (Table 9), each configuration's accuracy agrees with the RTX 5090's within the noise of Table 1 (B200 − RTX 5090: BF16 +1.9, W4A4 trace −2.8, MR-GPTQ +2.8; the exception, RTN, traces to one RTX 5090 run, Section 5.5), teacher-forced on the same BF16 traces all 6 checkpoints have the same per-token error on both kernels to within 0.0004 nats, trace calibration again shortens the traces (0.94× the tokens of chat GPTQ over all draws). The KV cache behaves differently. On SM100, vLLM quantizes queries to FP8 whenever the cache is quantized. With 16 samples per problem, FP8 KV costs no measurable accuracy (−1.0 [−3.3, +1.2] against BF16 KV on 8B W4A4 trace; equivalent within 2.8 points), but NVFP4 KV, which costs none at 8B on SM120, costs −5.9 [−8.8, −3.2] at 1.09× the tokens (14B, 4 samples: −1.4 [−5.3, +2.8]), so the KV format has to be checked per model and per kernel. With 180 GB the cache binds only for a BF16 cache at 128 such requests (Figure 5b). Limited to the RTX 5090's memory budget, it preempts at $C = 64$ in every configuration, and its throughput ratios follow the cache-size ratios within 12% (FP4 over FP8 weights with FP8 KV: 1.10×, cache 1.19×).

7 Discussion and Limitations

What the noise changes. At the scale of public math benchmarks, accuracy differences between FP4 configurations are resolved to a few points at best, while length and memory are measured precisely by one run and enter the serving rate directly; most of the differences we can establish are of the second kind. Single four-sample runs have tails as large as the gaps between published recipes (Sections 3 and 5.5), so rankings built on them need repeated runs. Table 4 lists the choices the evidence supports. Which activation statistics of thinking-mode text matter to GPTQ is open.

Table 4. What to do when serving a reasoning model with FP4 on Blackwell, and the evidence behind each choice.

Choice	Evidence
Calibrate GPTQ on reasoning traces (own or public)	removes most of chat calibration's extra length on Qwen3 in every draw, on both GPUs; accuracy gain within the noise (Section 5.1)
W4A4, not W4A16, when batches reach 64	W4A16 computes in BF16 (Section 6.1)
No residual rows on trace weights	activations equivalent to none within 2.4 points; the 8B FP4 cost is the weights' (Sections 4.1 and 5.1)
On a cache-bound card, always an FP8 KV cache	doubles concurrency; equivalent to BF16 KV within 2.8 points (Sections 6.2 and 6.5)
NVFP4 KV only after a per-model, per-kernel check	collapses 1.7B; costs 8B on SM100 (Sections 5.4 and 6.5)
FP4 weights only to fit a larger model whose lead exceeds its FP4 cost	at 8B, FP4 costs −3.9 points against FP8 + FP8 KV for 1.05× the rate; 14B in FP4 runs at the 8B FP8 rate but loses most of its AIME lead (Sections 6.2 and 6.4)
Compare PTQ methods over draws, $n \geq 16$, paired, past the cap	draw spread and sampling noise of Table 1

Limitations. We study Qwen3 at three sizes and one Llama-based distilled model, the latter for calibration only; the end-to-end evaluation is competition mathematics, with GPQA-Diamond, IFEval and MMLU-Pro as checks in other domains, and code generation is untested. The capacity model is fitted on one consumer GPU and assumes a full cache; it does not cover multi-GPU serving, prefix sharing or speculative decoding. On the B200 we test only its cache-ratio consequence. With 30 problems per year, paired intervals stay several points wide even pooled; the 1.7B KV results cover one year, and the released QAT and ModelOpt checkpoints and 1.7B chat GPTQ are single checkpoints. Teacher forcing cannot see errors that compound after a resampled token.

Artifacts. We will release the calibration traces, every raw sample and all scripts, including the SM120 FP-Quant plugin, the planner and the build of every number here.

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AI USAGE DISCLOSURE

This manuscript was originally drafted by the authors. To improve clarity and fluency, we used GPT-6 for language translation and copy-editing. Its role was limited to refining the presentation of existing text; all scientific content, including the research ideas, system design, experiments, and analysis, was developed entirely by the authors. We have verified the accuracy of all edited text and remain fully responsible for the work.